revealed that there are 14 different finch species on the Galápagos islands. The differences in their beaks are marked and significant. For example, cactus finches have long, pointed beaks that are ideal for picking seeds out of cactus fruits, while ground finches have shorter, stouter beaks that are better suited for eating large seeds on the ground. Warbler finches have slender, sharp beaks, which are ideal for catching flying insects.

Darwin speculated that the finches were descended from a common ancestral finch that had reached the archipelago from the mainland of South America. He concluded that a variety of finch populations had evolved in different Galápagos habitats, each group adapted for a more or less specialist diet by a process that he would later call “natural selection.” Over time, the finch populations had become distinct species.

In the early 21st century, researchers at Harvard University uncovered new evidence of how this happens at a genetic level. Their findings, published in 2006, showed that a molecule called calmodulin regulates the genes involved in shaping birds’ beaks, and is found at higher levels in longer-beaked cactus finches than in shorter-beaked ground finches.